

Air University



Network Security (Lab)

Year 2026

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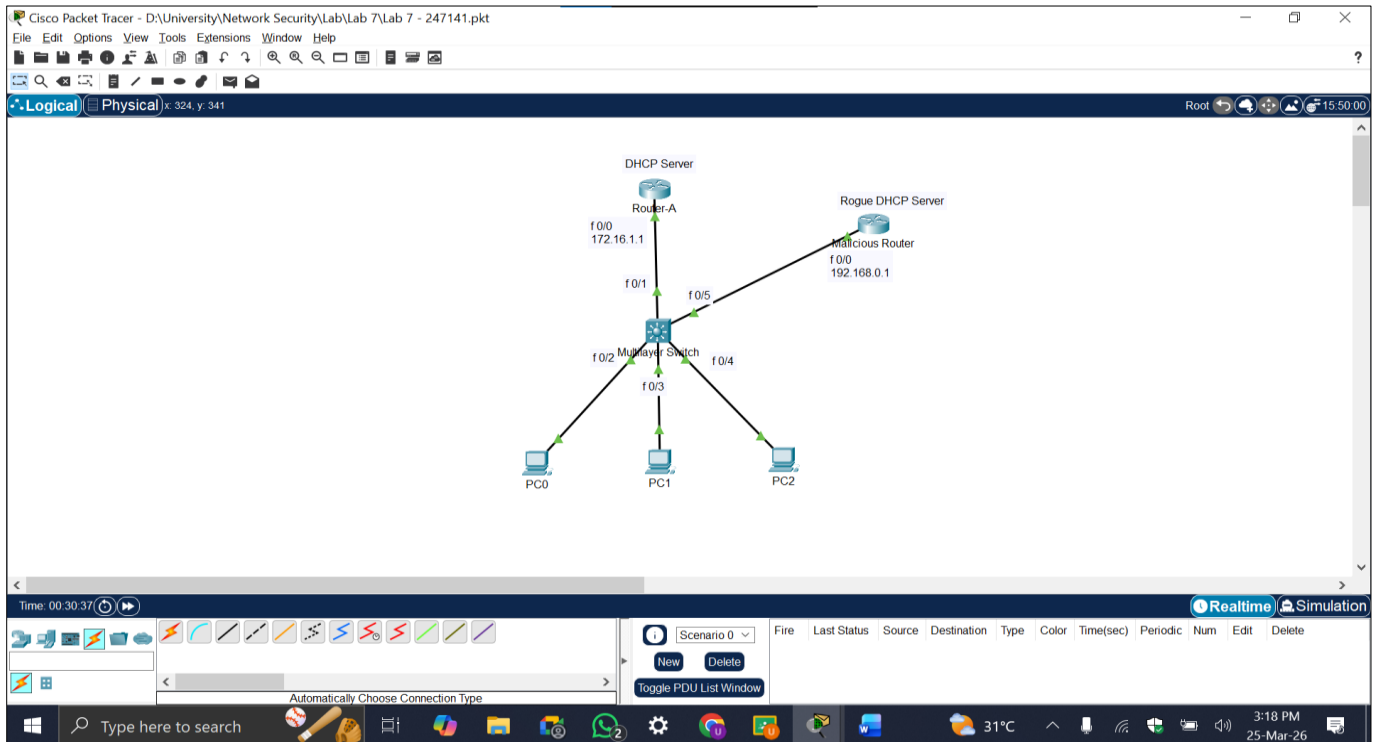
BS – Cyber Security – Fall – 24 (A)

Date of Submission: March 25, 2026

Lab # 07

DHCP Snooping

Topology:



Trusted DHCP Server Configuration:

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router-A
Router-A(config)#int f 0/0
      ^
% Invalid input detected at '^' marker.

Router-A(config)#int f 0/0
Router-A(config-if)#ip address 172.16.0.1 255.255.0.0
Router-A(config-if)#no shutdown

Router-A(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router-A(config-if)#exit
Router-A(config)#service dhcp
Router-A(config)#ip dhcp pool pool1
Router-A(dhcp-config)#network 172.16.0.0 255.255.0.0
Router-A(dhcp-config)#default-router 172.16.0.1
      ^
% Invalid input detected at '^' marker.

Router-A(dhcp-config)#default-router 172.16.0.1
Router-A(dhcp-config)#exit
Router-A(config)#ip dhcp excluded-address 172.16.0.1 172.16.0.10
Router-A(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

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Rogue DHCP Server Configuration:

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f 0/0
Router(config-if)#ip address 192.168.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#service dhcp
Router(config)#ip dhcp pool2
^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp pool pool2
Router(dhcp-config)#network 192.168.0.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.0.1
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.0.1 192.168.0.10
^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp excluded-address 192.168.0.1 192.168.0.10
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```

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DHCP Configuration on Switch:

```
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip dhcp snooping
Switch(config)#no ip dhcp snooping informtaion option
^
% Invalid input detected at '^' marker.

Switch(config)#no ip dhcp snooping informtaion option
^
% Invalid input detected at '^' marker.

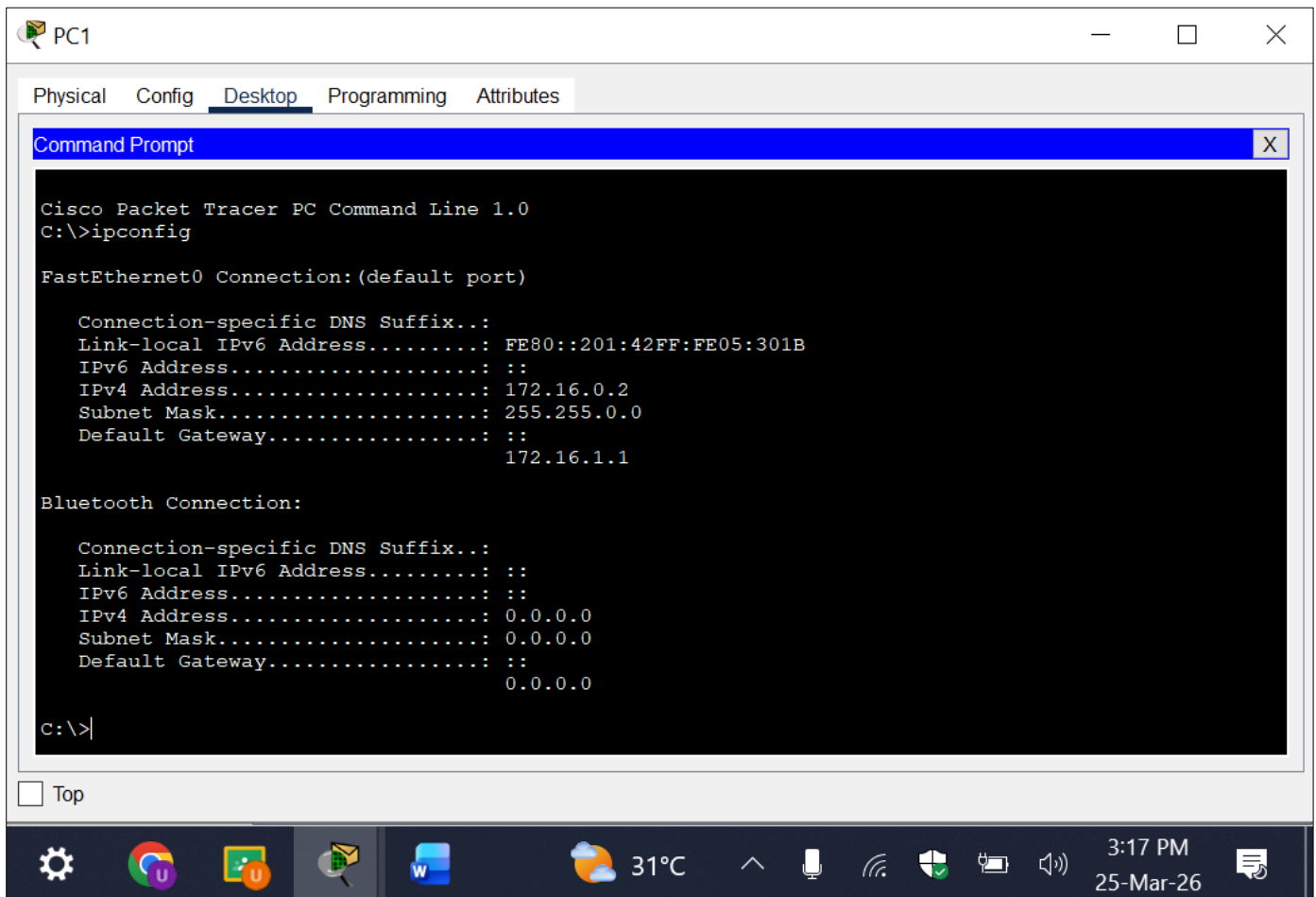
Switch(config)#no ip dhcp snooping information option
Switch(config)#ip dhcp snooping vlan 1
Switch(config)#int f 0/0
%Invalid interface type and number
Switch(config)#int f 0/4
Switch(config-if)#ip dhcp snooping trust
Switch(config-if)#show ip dhcp snooping
^
```

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Verification:

```
Switch#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
1
DHCP snooping is operational on following VLANs:
1
Smartlog is configured on following VLANs:
none
Smartlog is operational on following VLANs:
none
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is disabled
  circuit-id default format: vlan-mod-port
  remote-id: 00E0.F79A.4837 (MAC)
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:
```

Interface	Trusted	Allow option	Rate limit (pps)
FastEthernet0/3	no	no	unlimited
Custom circuit-ids:			
FastEthernet0/4	yes	yes	unlimited
Custom circuit-ids:			
FastEthernet0/1	no	no	unlimited
Custom circuit-ids:			
FastEthernet0/2	no	no	unlimited
Custom circuit-ids:			
FastEthernet0/5	no	no	unlimited
Custom circuit-ids:			

```
Switch#show ip dhcp snooping binding
```

MacAddress	IpAddress	Lease(sec)	Type	VLAN	Interface
00:07:EC:9A:14:AA	172.16.0.11	86400	dhcp-snooping	1	FastEthernet0/1
00:90:0C:68:AA:A1	172.16.0.12	86400	dhcp-snooping	1	FastEthernet0/2
00:90:2B:1A:55:19	172.16.0.13	86400	dhcp-snooping	1	FastEthernet0/3

```
Total number of bindings: 3
```

```
Switch#
```

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25-Mar-26

- End of Lab Task -